

PRD: AutoAssess AI Claims Automation Module

Version: 1.0

Status: Draft

Owner: Senior AI Product Manager

1. Vision & Goals

Vision: AutoAssess AI turns human agents into reviewers rather than manual estimate generators to enable faster, AI-driven damage assessments that result in faster payouts and higher accuracy, while focusing human expertise only where it adds material value.

Product Goals:

- Reduce the time from photo upload to repair authorization from days to minutes.
- Operational Efficiency by automating the review and assessment workflow.
- Standardization of repair estimates by utilizing consistent cost databases.

2. User Stories (Claims Agent)

As the primary user, the claims agent interacts with the tool to verify and finalize assessments. As a claims agent:

- I want to be able to view uploaded photos of damage for each claim.
- I want the system to automatically categorize and label damage in uploaded photos so that I can identify issues at a glance.
- I want a preliminary cost estimate generated for me so I can focus on reviewing rather than manual data entry so that I can save time.
- I want to be notified of low-confidence assessments so that I can prioritize complex cases that require human expertise.
- I want to easily override AI generated assessments so that I can ensure that the final report is accurate.
- I want my generated reports to be immediately accessible to the senior claims adjuster for review and approval so that the claim process can be completed quickly.
- I want the system to generate an authorization to proceed with repairs at an approved repair shop once the report has been approved by the claims adjuster.

3. Key Features

- **(P0) AI Damage Detection Engine:** A module that identifies scratches, dents, and structural damage from photos. It uses deep learning models to analyze pixels in uploaded photos, identifying specific damage signatures such as scratches, dents, or structural deformations

- **(P0) Automated Parts Mapping:** Automatically matches detected damage to specific vehicle components (e.g., front bumper, driver door, etc.). This is done using a deep learning image classification engine.
- **(P0) Dynamic Estimate Generator:** Integrates with repair cost databases to provide a line-item estimate for parts and labor. An image classification module assigns a severity score based on comparing with images from the repair cost database. The system queries the database to generate a repair cost estimate. A confidence score is assigned to each estimate, and if the score is below a certain threshold, the estimate is flagged for manual review by an agent.
- **(P0) Agent Verification Workspace:** A user interface for the agent to view the case including the photos and repair estimates. It allows the agent to view the images to verify the estimate, make changes, or approve the estimate. Where the AI generated estimate has a low confidence score, it is flagged for more detailed human review.
- **(P1) Report Approval and Repair Authorization:** The system also provides a workspace for the senior claims adjuster to review generated repair estimate reports, modify or approve them, and generate authorizations for repair.
- **(P1) Fraud Detection:** To prevent fraud or error, a verification module compares the photo against known information about the vehicle gathered during the initial claim report. The module uses OCR to compare the license plate number and VIN (Vehicle Identification Number) in the uploaded photos to the policyholder's registered vehicle. The make, model, and color are also assessed using a deep learning classification model to identify and confirm the make and model.

4. Feature Prioritization

- The damage detection, part mapping, estimate generation, and the agent workspace UI features are P0 priority features, because these represent the core value of the system.
- The report approval and repair authorization processes are prioritized as P1 features because while these are important features to complete the automation of the end-to-end workflow, they are aimed at a different user and can be added once the core product is built.
- The fraud detection features are important to maintain the integrity of the product, but they are not core features and can be delivered once the core product is built.

5. Prototype Scope (MVP)

In view of the above feature prioritization, the prototype that has been built is limited to the following features:

- Single-image upload
- Simulated damage detection output
- Confidence score + agent override

- Auto-generated repair summary (mocked)

5. Success Metrics (KPIs)

- **Average Claim Handling Time:** Reduction in time spent per claim by an agent.
- **Accuracy Rate:** Percentage of AI reports that are accepted by agents without edits.
- **First Contact Resolution:** Percentage of claims where an estimate is generated and authorized during the initial session without a need for additional steps.
- **Cost savings per claim:** Estimated based on reduction in human interventions required to process each claim.

6. AI Integration & Human-AI Interaction

The solution will leverage a multi-stage deep learning pipeline:

- **Image Classification:** To verify the photo is of the correct vehicle and angle, leveraging existing databases of vehicle photos to train an image classification model.
- **Damage Identification:** To highlight exact areas of damage, using a model that can precisely outline damaged areas of an auto part. This allows the system to estimate the size and extent of the damage.
- **Severity Analysis:** To determine if a part needs "Repair" vs. "Replacement" based on damage depth and location.

To maintain trust and accuracy, the system will employ a Human-in-the-Loop (HITL) model:

- **Confidence Scoring:** Assign a confidence percentage to every detection. Anything below 85% is flagged for manual review.
- **Seamless Overrides:** The UI is designed so that the human agent can easily modify the assessment and repair estimate. Every human edit is fed back into the model to improve future training cycles (Active Learning).
- **Transparency:** The system provides reasoning tags such as "*Detected structural crease, suggest replacement*") to help agents understand the automated logic.

7. Ethical Considerations

- **Bias Mitigation:** Ensure the AI models are trained on various makes, models, and lighting conditions to prevent unfair claim denials.
- **Data Privacy:** Ensuring all uploaded images are stripped of PII (personally identifiable information) before being used for model improvement.
- **Human accountability:** Final decisions regarding claims are made by humans to maintain accountability, and AI recommendations are auditable so they can be reviewed and assessed when needed.